

■ Accounts Receivable (AR) Module & Business Cycles

D365FO Middleware Backend — AR Domain, Processors & Invoicing Flows

Target Audience: Finance (AR/Billing), Logistics Operations, Software Engineers.

1. Business Domain Overview

The **Accounts Receivable (AR)** module manages customer revenue billing for all logistics operations:

- **Sea & Air Freight Forwarding Invoices:** Billing clients for ocean container freight, air cargo charges, and port handling.
- **Inland Fleet / Trucking Invoices:** Billing clients for land container haulage and trucking trips.
- **Yard & Terminal Services Invoices:** Billing clients for container storage, demurrage, and yard gate fees.
- **Shipping Line Charges:** Billing shipping lines or co-loaders for agency and handling services.
- **Credit Notes:** Issuing adjustment vouchers for invoice corrections or client discounts.

2. AR Entry Processors & Business Scope

Processor Class Name	EntryProcessorTypes	D365FO Target Entity	Real-World Business Context
AccountReceivableFreightEntryProcessor	AccountReceivableFreight	FreeTextInvoiceHeaders / Lines	Ocean & Air Freight Customer Invoices
AccountReceivableFreightCreditNoteEntryProcessor	AccountReceivableFreightCreditNote	FreeTextInvoiceHeaders / Lines	Freight Invoice Adjustment Credit Notes
AccountReceivableTruckingEntryProcessor	AccountReceivableTrucking	FreeTextInvoiceHeaders / Lines	Land Trucking & Transport Customer Invoices
AccountReceivableTruckingCreditNoteEntryProcessor	AccountReceivableTruckingCreditNote	FreeTextInvoiceHeaders / Lines	Trucking Invoice Credit Notes
AccountReceivableYardEntryProcessor	AccountReceivableYard	FreeTextInvoiceHeaders / Lines	Container Storage & Yard Fee Invoices
AccountReceivableShippingLinesEntryProcessor	AccountReceivableShippingLines	FreeTextInvoiceHeaders / Lines	Shipping Line Carrier Service Invoices

3. Financial & Accounting Impact in D365FO

When an AR processor posts a batch to D365FO as a **Free Text Invoice**, the following accounting entries are created:

Accounting Posting Entries:

- **DEBIT:** Customer Subledger Account (**CustAccount** e.g. **CUST-00192**)
- **CREDIT:** Revenue Main Account (e.g., **410101** Freight Revenue or **410201** Trucking Revenue)
- **CREDIT:** Sales Tax / VAT Payable Account (e.g. **220101** output VAT 14% if applicable)

Mandatory Financial Dimensions Enforced:

Each line item tagged on the Free Text Invoice header/line must contain valid financial dimensions:

- **MainAccount:** Revenue GL account code
- **BusinessUnit:** e.g. **BU-LOG** (Logistics)
- **CostCenters:** e.g. **CC-FRT** (Freight Ops) or **CC-FLEET** (Trucking Ops)
- **Activity:** Operational project code (e.g. **EXPORT-FCL**)
- **Location:** Branch code (e.g. **LOC-ALY**)
- **Customer:** Customer account code
- **SalesMan:** Salesperson ID responsible for commission tracking
- **CoordinatorMan:** Operations coordinator managing the shipment

4. End-to-End AR Business Cycle Flow

```
[Operational Manifest / Trip Sheet Completed in Excel]
■
▼
[API Endpoint: Upload AR Excel]
■
▼
[AccountReceivableFreightEntryProcessor]
(Parses Dimensions, Validates Customer, FX Rate)
■
▼
[DataBatch Status: PENDING (Max 1000 Lines)]
■
▼
[BullMQ Queue: dfo-free-text-invoice-queue]
■
▼
[FreeTextInvoicePostingStrategy]
■
▼
[Post OData to D365FO FreeTextInvoiceHeaders]
■
▼
[D365FO Creates Posted Free Text Invoice]
(Generates Invoice Number & Debits Customer Subledger)
■
▼
[Batch Status Set to COMPLETED]
```

Detailed Step-by-Step Cycle Execution:

1. **Invoice Excel Upload:** Operations staff uploads customer billing sheets containing customer account numbers, bill of lading (B/L) numbers, charge descriptions, and amounts.
2. **Dimension Extraction:** The processor extracts dimensions from **ACCOUNTDISPLAYVALUE**.
3. **Customer & Tax Lookup:** The processor validates the customer account against D365FO master data cache and applies the appropriate **SALESTAXGROUP** (e.g., **VAT_14**).
4. **Exchange Rate Application:** For foreign currency billing (e.g., USD or EUR), transaction date exchange rates are retrieved from D365FO **ExchangeRates** service to calculate reporting currency equivalent in EGP.
5. **Queue Posting:** BullMQ dispatches the job to **dfo-free-text-invoice-queue**.
6. **OData Posting:** The strategy creates invoice headers in **FreeTextInvoiceHeaders** and line items in **FreeTextInvoiceLines**.
7. **Confirmation:** D365FO returns the assigned Invoice Number (e.g., **FTI-2026-00912**), updated in MongoDB batch records.